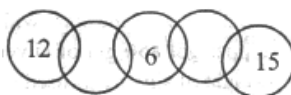


JMO – 2017

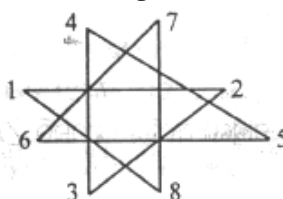
Answer all questions. All questions carry equal marks.

Full Marks 100

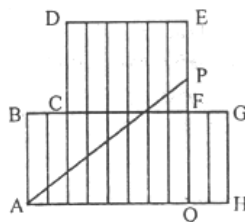
1. In the given figure a total of 9 regions are formed by 5 circles. Three such regions represented by number 12, 6 and 15. Use any other number once only for the rest of the regions, so that sum of the numbers in each of the circle is 20.



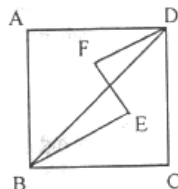
2. Evaluate: $\frac{1}{3 \times 7 \times 11} + \frac{1}{7 \times 11 \times 15} + \frac{1}{11 \times 15 \times 18} + \dots$ upto 20 terms.
3. In the given figure find the sum of the vertex angles.



4. In the given figure ABCDEFGH is an octagon comprising of 16 congruent rectangles of size 4 X 1 each. P is a point on EF such that AP divides the octagon into two parts of equal area. Find the length of AP.



5. Find the largest positive integer n so that $n^3 + 100$ is exactly divisible by $n + 10$.
6. A packet has less than 100 chocolates which when equally distributed to 7, 8 or 9 children 2, 5 and 3 chocolates are left out. What should be the maximum number of children so that the chocolates can be equally distributed among them with no chocolates left out and each of them will get more than 1 chocolate?
7. In the Mathematics Olympiad Exam small silver stars were awarded top distinct scorers. All stars are of same weight. One gets the number of stars which is half of the number of stars with his/her predecessor. The lowest ranker got 1 star and the toper got 850 gm of silver. If one star was kept in the Institute as sample then what weight of silver was used in making all the stars?
8. In the square ABCD, E and F are the points on the opposite side of the diagonal BD, $\angle BEF = \angle EFD = 90^\circ$, $BE = 14$, $EF = 3$ and $FD = 7$ units, find the area of the square.



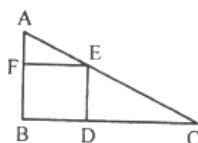
9. Three balls A, B, C are coloured with three different colours such as red, green and white. If out of the following only one statement is true then find the colour of each ball?
(a) A is red, (b) B is not red (c) C is not green

JMO QUESTIONS (ORISSA)

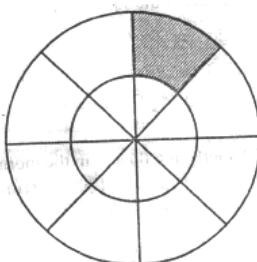
10. The sum of digits of a number $10^n - 7$ is 3792 then what is the value of n ?
11. In the giving addition, each letter represents a different digit. What is the missing digit?

$$\begin{array}{rcccc} & A & B & C & D \\ + & & B & C & D \\ \hline G & H & I & J & K \end{array}$$

12. In the given figure ABC is a right triangle with $\angle B = 90^\circ$, $BC = 8$ cm, $AB = 4$ cm. BDEF is a square inscribed in it. Find the side of the square.



13. Exactly at 6.00 AM in the morning the measure of angle between the hour hand and the minute hand was 180° . Determine the exactly after what time again the measure of the angle between the two hands will be 180° ?
14. 'abc' represents a three digit number where a, b, c are the digits of it and 'bc' also represents a two digit number where b, c are digits of it. If $abc = (bc)^2$ and $bc = (c)^2$, then which digits are represented by a, b, c ?
15. Determine the smallest counting number having 36 factors with 6 consecutive factors.
16.
(a) If one purchase 4 lemons for Rs. 3 and sells 3 lemons for Rs. 4, determine the % of profit.
(b) Find the largest remainder when a two digit number is divided by the sum of its digits.
17. Determine the remainder when 7^{2018} is divided by 2017.
18. A piece of paper from an exam answer sheet has been torn. If the sum of pages of the rest papers of this answer sheet 40 then determine the total pages of the answer sheet and the page number of the torn piece of paper.
19. In the given figure a circle of radius 28 cm is divided into 8 equal parts. Taking the same centre another circle of radius 14 cm has been drawn. Find the area of the shaded portion.



20. The distance between two places A and B along the road joining them is 240 KM. Two persons P and Q started at the same time from A and B respectively and continues driving their car towards each other. P's speed was double the speed of Q. After meeting each other, they returned to their starting place immediately. But while returning P's speed was half of his former speed and Q's speed was double of his former speed. Soon after reaching their starting place they again moved to meet each other. On this occasion the speed each of them was equal to their respective first journeys. If the time interval between their first and second meeting was 5 hours, determine the initial speed of the P and Q.
